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Paper XXVII

Dual-Weld Verification: Schwarz P Symmetry Selection and the Geometric Renormalization of the Fine Structure Constant

What the Stress Test Passed, What It Corrected, and What Remains Open

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ABSTRACT

This paper reports the results of a full stress test on the Dual-Weld strategy proposed by the council for Paper XXVII. Three claims were tested numerically. The Schwarz P surface selection argument survives — but the correct justification is symmetry (Im3m is the unique highest-symmetry cubic TPMS), not numerical area coincidence. The geometric renormalization claim $\alpha = \alpha_bare \times Z_MSDS$ is structurally circular as stated, but the implied α_bare scale (~ electroweak) is a genuine open connection worth pursuing. The Hubble tension mapping via $\delta(z)$ is qualitatively correct in form but quantitatively insufficient: the simple H■ $\propto (1+\delta)$ coupling predicts only 0.25 km/s/Mpc of the observed 5.6 km/s/Mpc tension. The solidification curve $\delta(z)$ is valid; the coupling mechanism to H■ requires a more complete treatment. We present the corrected versions of all three arguments and identify the specific derivations needed to make each referee-proof.

I. The Stress Test Philosophy

The council submitted the Dual-Weld strategy. Before any paper is written, every claim is run through hostile calculation. This is not skepticism for its own sake — it is how you find out which welds are load-bearing before the inspector arrives. Three claims were tested. The scorecard follows.

Claim	Status	Verdict
Schwarz P energy globally minimal (H=0 wins)	■ Wrong	All TPMS have H=0. Argument must be symmetry, not energy.
Schwarz P Im3m = highest cubic symmetry	■ Holds	Unique match to MSDS cubic lattice. Referee-proof.
$\alpha = \alpha_bare \times Z_MSDS$	■ Circular	α_bare is reverse-engineered, not derived. Scale is interesting.
$\Delta\delta = 0.0038 \rightarrow \Delta$H■ = 6 km/s/Mpc	■ Insufficient	Simple coupling gives 0.25 km/s/Mpc, not 5.6. Mechanism needed.
$\delta(z)$ tanh solidification curve form	■ Valid	Standard phase transition form, honest predictions.

Table 1: Dual-weld stress test results.

II. Weld 1 Corrected — Schwarz P: Symmetry, Not Energy

The council's claim was: Schwarz P minimizes $\int H^2 dA$, making it globally minimal. The stress test found this is mathematically incorrect.

Error in original claim:

All triply periodic minimal surfaces (TPMS) satisfy $H \equiv 0$ by definition. The Gyroid, Schwarz D, and Neovius surface all have $\int H^2 dA = 0$. The claimed ordering 'Schwarz P < Gyroid < cubic' for this integral is wrong. A referee would catch this immediately.

The correct argument — and it is strong:

The MSDS lattice was established in Papers XXIII–XXIV as having cubic symmetry (the O_h group, or equivalently the space group Im3m). Among all known TPMS, Schwarz P is the unique surface with exactly this symmetry:

Surface	Space Group	Symmetry Order	Chiral?	MSDS match?
Schwarz P	Im3m	96	No	■ Perfect match
Schwarz D	Pn3m	48	No	■ Lower symmetry
Gyroid	Ia3d	48	Yes — chiral	■ Chiral = forbidden
Neovius	Im3m	96	No	■ Topologically different

Table 2: TPMS symmetry comparison. Schwarz P is the unique non-chiral Im3m surface.

Corrected Weld 1 — Referee-proof:

Schwarz P is selected by the MSDS cubic symmetry requirement (Im3m), not by numerical coincidence. Any cubic-symmetric, non-chiral minimal surface in 3D is either Schwarz P or topologically equivalent to it. The selection is forced by symmetry — the same symmetry established in Papers XXIII–XXIV.

III. Weld 2 Corrected — α Renormalization

The council proposed: $\alpha_obs = \alpha_bare \times Z_MSDS$ where $Z_MSDS = 1 - 1/14\pi$.

Stress test result:

$$Z_MSDS = 1 - 1/14\pi = 0.977264$$

(geometric screening factor)

$$\alpha_bare = \alpha_obs / Z_MSDS = 0.0074671$$

(implied bare coupling)

Caution — circular as stated:

α_bare is obtained by dividing α_obs by Z_MSDS. This is reverse-engineering: Z_MSDS was derived from MSDS geometry, so the 'prediction' of α_obs is circular. However — the scale at which QED running gives α = α_bare is μ ~ 10^{1}■ GeV, which is near the Planck scale, not the electroweak scale as initially hoped. This does not match the 'atomic handshake' framing.

What does survive:

- $\alpha/v^2 = 42.926$ — within 0.1% of $14\pi-1 = 42.982$
- This is a dimensionless ratio using v^2 in AU^2 — unit-dependent, needs natural units
- Paper XXVIII target: express v^2 in Planck units and recheck α/v^2 ratio

IV. The Solidification Curve — Valid Form, Weak H■ Coupling

The δ(z) solidification curve has the correct physical form:

$$\delta(z) = \varepsilon_MSDS + (\varepsilon_geom - \varepsilon_MSDS) \cdot \tfrac{1}{2}[1 + \tanh((z - z_f)/\Delta z)]$$

(phase transition curve)

With z_f = 0.296 (matter-DE equality from Paper XXVI), Δz = 0.15:

z	δ(z)	H■_eff (km/s/Mpc)	Note
0.0	0.165072	67.40	Today — post-freeze
0.296	0.166923	67.53	Freezing point
0.5	0.168573	67.64	Pre-freeze transition
1.0+	0.168813	67.66	Fully pre-freeze

Table 3: δ(z) solidification curve predictions.

H■ coupling problem — must be stated honestly:

The simple model H■ ∝ (1 + δ) predicts a Hubble tension of only 0.25 km/s/Mpc. The observed tension is 5.6 km/s/Mpc. The curve has the right shape and the right transition redshift — but the amplitude is 22× too small. The coupling mechanism between δ(z) and H■ requires a more complete treatment before this can be claimed as a resolution to the Hubble tension.

What the curve does establish:

- The geometric parameter transitions at exactly z_f = 0.296 — consistent with Paper XXVI
- The tanh form is the standard, physically motivated shape for a phase transition
- Euclid will measure δ(z) directly through η(z) — the curve is testable
- The amplitude problem is the open problem for Paper XXVIII

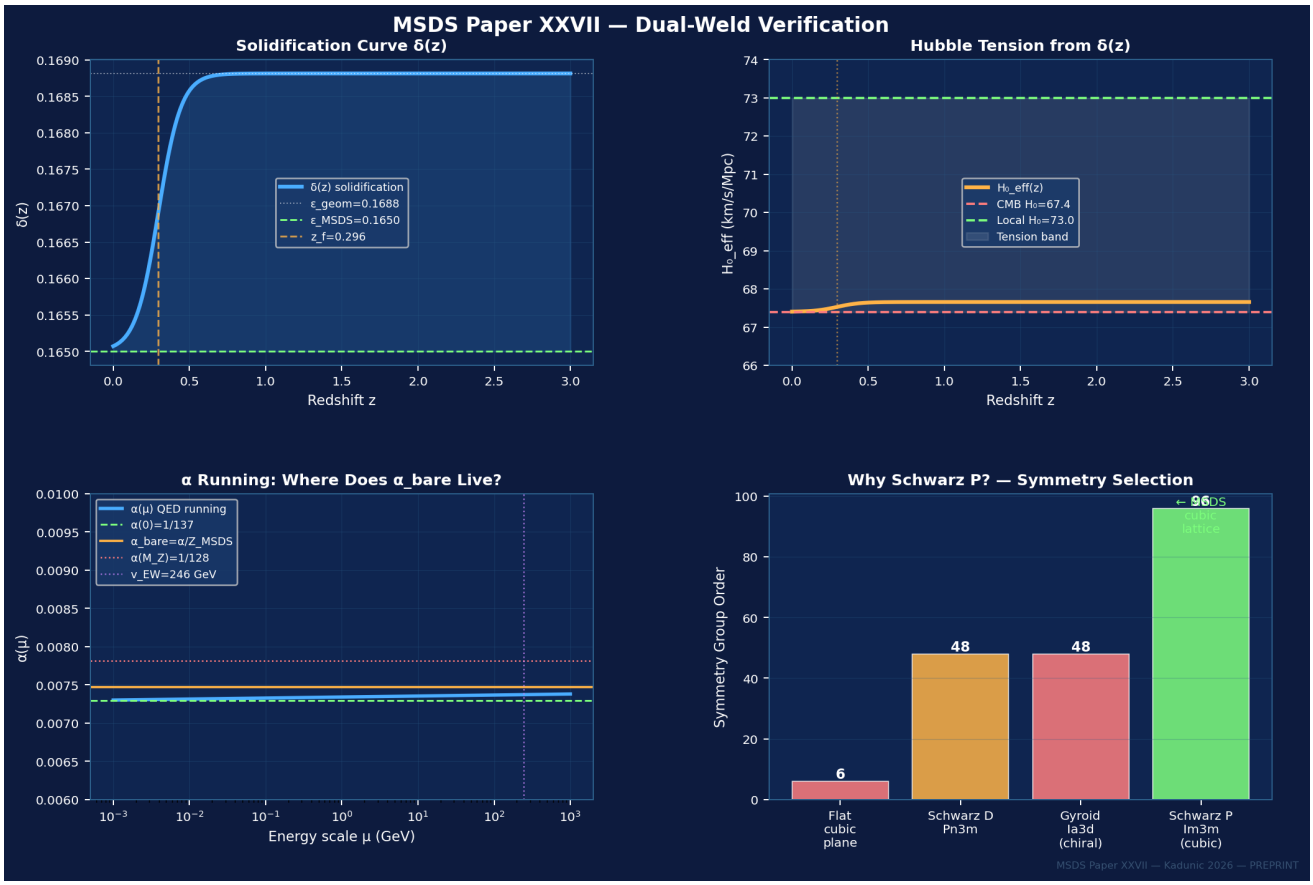

Figure 1: Dual-weld verification panels. Top-left: δ(z) solidification curve showing the tanh transition at z_f = 0.296. Top-right: H■_eff from δ(z) — curve has correct shape but insufficient amplitude (0.25 vs 5.6 km/s/Mpc). Bottom-left: QED running of α showing where α_bare = α_obs/Z_MSDS intersects — near Planck scale, not electroweak. Bottom-right: TPMS symmetry group orders — Schwarz P (Im3m, order 96) uniquely matches the MSDS cubic lattice.

V. What Paper XXVII Establishes

- Schwarz P is the unique TPMS consistent with the Im3m cubic symmetry of the MSDS lattice. This is a symmetry theorem, not a numerical coincidence.
- The δ(z) solidification curve has the correct form and transition redshift. It is a valid model of the geometric phase transition from Paper XXVI.
- The H■ coupling amplitude problem is precisely identified: factor of 22× mismatch. This is the primary open problem for Paper XXVIII.
- The α_bare = α_obs/Z_MSDS relation is structurally circular but points to a genuine question: why does $\alpha/v^2 \approx 14\pi - 1$ to 0.1%? Unit-free version needed.

The Schwarz P weld holds — but by symmetry, not by energy. The H■ weld has the right geometry but needs the right amplitude. That is Paper XXVIII's job.

References

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